

Week 21– Rebound Height Experiment

Grade: Grade 9 Unit: Linear Relations

Curriculum Expectations

MPM 1D/ MFM 1P: Pose problems, identify variables, and formulate hypotheses associated with relationships between two variables

SEL: Make connections between math and everyday contexts to help them make informed judgements and decisions

Activity

- 1) Students will conduct an investigation where they drop a ball from various heights and determine the resulting rebound height.
 - 2) Students will find a ball of their choice and find an open area to complete this activity.
 - 3) Before starting the investigation students will make a hypothesis of how the height that the ball is dropped from will impact the rebound height, along with the maximum and minimum rebound height of the ball.
 - 4) After they have made their hypothesis students will complete the investigation and fill in the chart as they go along.
 - 5) Students may wish to use a measuring tape, ruler, metre stick, etc. or can use nonstandard units of measurement if they don't have access to a measuring device.
 - 6) After the investigation students will display their data in a graph.
- Challenge- Try the investigation with a different type of ball. Does the first or second ball go higher? Why do you think that one ball went higher than the other?

Check for Understanding

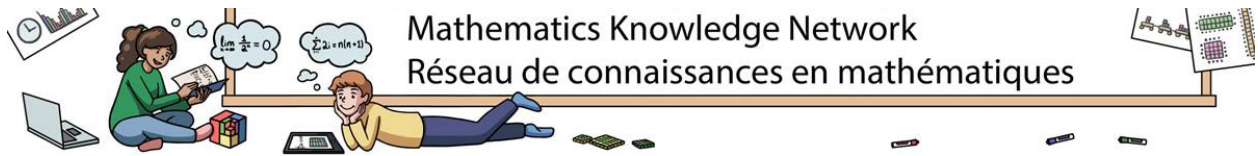
I can make a hypothesis about the relationship between the rebound height of a ball and the height from which it was dropped.

I can carry out an investigation to test my hypothesis.

I can display my findings from the investigation in a graph.

Materials

Recording sheet and graph paper attached below, pencil, measuring device, ball, and an open space



Dropped Height	Rebound Height

